

A large version of the SKEIN ENGINE logo, with "SKEIN" in black and "ENGINE" in orange, flanked by vertical orange lines.

ENGINE CONSTRUCTION MASTER VOLUME PLAN

Replacement implementation manuals for the current SKEIN Engine architecture

DOCUMENT CODE	SKEIN-IMP-002-MASTER
VERSION	0.1
DATE	30 July 2026
STATUS	CONSTRUCTION SET BLUEPRINT
TARGET SET	12 volumes / 2,185 planned pages

PURPOSE

This document locks the architecture, chapter inventory, dependency order and page budget for the full replacement of SKEIN-IMP-001. It is the authoring contract for the new construction manuals, not an implementation volume itself.

0. Document mandate

The SKEIN-IMP-002 construction set will replace the historical SKEIN-IMP-001 volumes as the active coding reference. The new set is based on SKEIN-SPEC-001 v0.4 and the live Windows x64 CMake project. Historical material may inform coverage, but no historical decision survives automatically.

LOCKED SCALE

The replacement set must contain at least 1,500 substantive implementation pages. The present plan budgets 2,185 pages across twelve volumes so that necessary detail is not squeezed into brochure-sized summaries.

0.1 Source-of-truth hierarchy

SKEIN-SPEC-001 v0.4 defines the current product and architectural scope.

SKEIN-IMP-002-MASTER defines the construction volume boundaries and authoring order.

Individual SKEIN-IMP-002 volumes define implementation contracts and completion evidence.

Accepted architecture decision records may refine a volume without silently broadening product scope.

The current repository and verified build behaviour override stale examples in older documents.

0.2 Replacement policy

SKEIN-IMP-001 remains archived as a historical reference.

SKEIN-IMP-002 uses a new document code and does not overwrite historical PDFs.

A new volume becomes authoritative only after its review gate is passed.

Until a replacement volume is approved, the master specification and current source tree remain authoritative.

Voxel-specific construction material is omitted unless voxels return to the approved master specification.

0.3 Volume writing standard

Every chapter states purpose, boundary, dependencies, public contracts, ownership, lifetime, threading, allocation, diagnostics, tests, performance expectations and completion evidence.

Code examples must compile conceptually against the documented module boundaries and use the four supported presets.

Third-party APIs are contained behind SKEIN-owned interfaces.

Examples distinguish normative contracts from illustrative implementation detail.

Page count must come from substantive engineering content, not duplicated boilerplate or decorative padding.

1. Construction set overview

VOL	TITLE	PAGES	CHAPTERS
01	Core Systems and Runtime Foundation	240	15
02	Reflection, Serialization and Object Model	165	14
03	Mathematics, Scenes, Worlds and Large Coordinates	180	14
04	RHI, Vulkan and Direct3D 12	215	15
05	Forward+, Deferred Rendering and Render Features	225	15
06	Shaders, Materials, Assets and Content Pipeline	175	15
07	Physics, Animation, Audio, Input and Navigation	190	15
08	SkeinScript and Plugin Architecture	155	14
09	SkeinUI, SkeinEditor and Project Framework	210	18
10	Networking and Starcore Server Architecture	145	13
11	SkeinInsights, Diagnostics and Profiling	135	15
12	Build, Testing, Deployment and Governance	150	14
PLANNED TOTAL		2,185 pages across 12 volumes and 177 chapters. A controlled variance of +/- 10 percent is acceptable if chapter obligations remain complete.	

1.1 Major dependency spine

Volume 01 establishes the buildable runtime foundation required by every later volume.

Volume 02 supplies reflection and serialization contracts consumed by scenes, assets, scripting, plugins and the editor.

Volume 03 establishes world, component and large-coordinate contracts consumed by rendering, physics and networking.

Volumes 04 and 05 establish the graphics foundation and renderer paths.

Volume 06 builds the content pipeline over reflection, serialization and RHI contracts.

Volumes 07 and 08 add simulation, gameplay scripting and extension surfaces.

Volume 09 builds SkeinUI and SkeinEditor over the prior runtime and content systems.

Volumes 10 and 11 add networking, Starcore and deep observability.

Volume 12 governs verification, deployment and release evidence across the entire set.

1.2 Recommended implementation waves

WAVE	VOLUMES	OUTCOME
Wave A	01, early 11, early 12	Foundation, logging, tracing hooks, tests and CI.
Wave B	02 and 03	Reflection, serialization, object model, scene and large-world contracts.
Wave C	04 and 05	RHI backends, render graph, Forward+ and deferred renderer.
Wave D	06 and 07	Assets, materials, Jolt, animation, audio, input and navigation.
Wave E	08 and 09	SkeinScript, plugins, SkeinUI and SkeinEditor.
Wave F	10 and full 11	Networking, Starcore and complete SkeinInsights analysis.
Wave G	12 completion	Packaging, release gates and governance closure.

2. Mandatory chapter anatomy

A. Purpose and boundary

The reason the subsystem exists, the problem it solves, explicit exclusions and the owning module.

B. Dependency gate

What must already compile, which contracts must be approved and which third-party decisions must be complete.

C. Public contract

Headers, types, functions, handles, events, versioning and error semantics exposed outside the module.

D. Internal architecture

Private implementation structure, algorithms, storage, scheduling and backend adapters.

E. Ownership and lifetime

Who owns each object, destruction thread, weak references, cancellation and shutdown behaviour.

F. Threading model

Thread affinity, synchronization, lock ordering, job boundaries and data handoff rules.

G. Memory model

Allocation source, expected volume, transient versus persistent data, budgets and instrumentation.

H. Diagnostics

Logging categories, assertions, counters, SkeinTrace channels, debug visualizations and failure messages.

I. Implementation sequence

Ordered work packages with intermediate compile and test gates.

J. Verification

Unit, integration, stress, performance, dual-backend or dual-compiler evidence as appropriate.

K. Completion gate

Observable criteria that permit the next dependent chapter to begin.

L. Deferred register

Deliberately excluded work and the condition that would permit reconsideration.

3. Cross-cutting engineering rules

RULE	MANDATE
Compiler parity	Every committed milestone must configure, compile and test under msvc-debug, msvc-release, clang-debug and clang-release.
Warnings	Warnings are errors. Compiler-specific options are gated by compiler identity, not merely MSVC-compatible command syntax.
Platform isolation	Win32 and future Linux APIs remain inside platform or backend modules.
Third-party containment	Jolt, Vulkan, D3D12, DXC, font libraries and other dependencies do not leak into unrelated public interfaces.
Large worlds	Any system storing or interpreting position must document world-space precision and local-space conversion.
Tracing	Hot systems define trace channels and measurable overhead budgets from their first implementation.
Shutdown	Every async operation documents cancellation and shutdown ordering.
Plugins	Public extension points are versioned; internal classes are not accidental plugin APIs.
No hidden globals	Global state requires explicit ownership, startup phase and test reset behaviour.
Evidence	Completion includes retained test output, benchmark results or captures, not only a successful demonstration.

3.1 Page-budget interpretation

The chapter budgets below are planning figures. They include diagrams, interface tables, code examples, test matrices and implementation checklists. They exclude blank separators and intentionally sparse title pages. Authors may move pages between chapters within a volume, but material movement between volumes requires an update to this master plan.

Volume 01 - Core Systems and Runtime Foundation

Repository, toolchain, foundation library, containers, memory, platform, jobs, lifecycle, configuration and core diagnostics.

VOLUME BUDGET

240 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V01.

Volume outcomes

Lock the public contracts and implementation order for core systems and runtime foundation.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
01.01	Construction rules and architectural invariants	12
01.02	Repository and module topology	14
01.03	CMake and four-preset toolchain	18
01.04	Build configuration and feature switches	12
01.05	Foundation types and numeric conventions	16
01.06	Error, result and assertion model	16
01.07	Strings, paths, hashing and identifiers	18
01.08	Container library policy	20
01.09	Memory architecture	34
01.10	Platform abstraction - Windows x64	24
01.11	Synchronization primitives	16
01.12	Job system and worker scheduler	30
01.13	Engine lifecycle and service registry	20
01.14	Core logging, CVars and command line	14

ID	CHAPTER	PAGES

01.15	Volume acceptance programme	10
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Chapter detail budget totals 274 pages. The remaining -34 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 01 detailed chapter inventory

01.01 Construction rules and architectural invariants [12 pages]

Coverage: Scope control; public/private boundaries; ownership vocabulary; thread-affinity notation; evidence retention.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.02 Repository and module topology [14 pages]

Coverage: directory standard; module naming; public/private includes; dependency rules; generated code boundaries.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.03 CMake and four-preset toolchain [18 pages]

Coverage: MSVC Debug/Release; clang-cl Debug/Release; Ninja; compiler feature probes; warning policy; build reproducibility.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.04 Build configuration and feature switches [12 pages]

Coverage: build macros; runtime feature flags; shipping restrictions; configuration files; version stamping.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.05 Foundation types and numeric conventions [16 pages]

Coverage: fixed-width types; byte and span types; strong IDs; endianness; units and durations.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.06 Error, result and assertion model [16 pages]

Coverage: Result<T>; error codes; assert tiers; fatal handling; noexcept policy; failure injection.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.07 Strings, paths, hashing and identifiers [18 pages]

Coverage: UTF-8 policy; StringView; path normalization; UUIDs; stable hashes; name table.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.08 Container library policy [20 pages]

Coverage: standard-library use; small vectors; flat maps; ring buffers; intrusive lists; iterator invalidation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.09 Memory architecture [34 pages]

Coverage: allocator interface; tags; arenas; pools; frame allocators; TLS scratch; budgets; leaks; fragmentation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.10 Platform abstraction - Windows x64 [24 pages]

Coverage: process; windows; clock; virtual memory; threads; dynamic libraries; file dialogs; crash dumps.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.11 Synchronization primitives [16 pages]

Coverage: mutexes; RW locks; events; semaphores; atomics; lock ordering; contention metrics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.12 Job system and worker scheduler [30 pages]

Coverage: work stealing; job graph; priorities; barriers; parallel-for; affinity; shutdown; profiling.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.13 Engine lifecycle and service registry [20 pages]

Coverage: startup phases; module graph; tick phases; shutdown; restartability; headless mode.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.14 Core logging, CVars and command line [14 pages]

Coverage: structured logs; sinks; categories; CVars; commands; configuration layering.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

01.15 Volume acceptance programme [10 pages]

Coverage: unit tests; stress tests; dual-compiler gate; performance budgets; completion evidence.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 01 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

This is the root construction volume. Early tracing and test hooks are coordinated with Volumes 11 and 12 but do not wait for their full applications.

Volume 02 - Reflection, Serialization and Object Model

Generated reflection, metadata, object handles, archives, schema evolution, transactions, prefabs and persistence contracts.

VOLUME BUDGET

165 planned pages across 14 chapters. Document code: SKEIN-IMP-002-V02.

Volume outcomes

Lock the public contracts and implementation order for reflection, serialization and object model.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
02.01	Reflection goals and non-goals	10
02.02	Metadata declaration syntax	14
02.03	Reflection code generator	20
02.04	Type registry and stable type IDs	16
02.05	Property and function descriptors	16
02.06	Object handles and lifetime tracking	18
02.07	Archive abstraction	12
02.08	Binary serialization format	16
02.09	Human-readable project formats	12
02.10	Schema versions and migration	14
02.11	Reference serialization	12
02.12	Transactions and undo/redo	14
02.13	Prefab data contracts	12
02.14	Persistence tests and completion gate	9

Chapter detail budget totals 195 pages. The remaining -30 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 02 detailed chapter inventory

02.01 Reflection goals and non-goals [10 pages]

Coverage: editor needs; script needs; serialization needs; plugin needs; no unrestricted RTTI.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.02 Metadata declaration syntax [14 pages]

Coverage: annotations; macros versus attributes; generated headers; source locations; diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.03 Reflection code generator [20 pages]

Coverage: scanner; AST extraction; incremental generation; CMake hooks; deterministic output.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.04 Type registry and stable type IDs [16 pages]

Coverage: registration; module ownership; inheritance; interfaces; hot reload conflicts.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.05 Property and function descriptors [16 pages]

Coverage: accessors; attributes; ranges; defaults; editor flags; script exposure.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.06 Object handles and lifetime tracking [18 pages]

Coverage: generation handles; weak handles; deferred destruction; object registry; thread rules.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.07 Archive abstraction [12 pages]

Coverage: reader/writer; binary/text; endianness; limits; error propagation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.08 Binary serialization format [16 pages]

Coverage: headers; type tables; field IDs; compression hooks; checksums; unknown fields.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.09 Human-readable project formats [12 pages]

Coverage: diff friendliness; canonical ordering; comments policy; round trips.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.10 Schema versions and migration [14 pages]

Coverage: version graph; migrators; deprecated fields; compatibility tests; rollback.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.11 Reference serialization [12 pages]

Coverage: asset refs; object refs; cross-scene refs; soft refs; missing targets.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.12 Transactions and undo/redo [14 pages]

Coverage: command records; object deltas; merging; memory budgets; plugin operations.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.13 Prefab data contracts [12 pages]

Coverage: base prefab; overrides; nested instances; variants; conflict diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

02.14 Persistence tests and completion gate [9 pages]

Coverage: golden files; fuzzing; corruption; migration matrix; performance.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 02 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 02 consumes approved contracts from earlier volumes, especially Volume 01 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 03 - Mathematics, Scenes, Worlds and Large Coordinates

Mathematics, transforms, scene graph, components, spatial indexing, world cells, streaming and precision-safe large worlds.

VOLUME BUDGET

180 planned pages across 14 chapters. Document code: SKEIN-IMP-002-V03.

Volume outcomes

Lock the public contracts and implementation order for mathematics, scenes, worlds and large coordinates.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
03.01	Mathematics conventions	12
03.02	Vectors, matrices and quaternions	16
03.03	Transforms and hierarchy propagation	18
03.04	Bounds and geometric primitives	12
03.05	Double-precision world coordinates	18
03.06	Camera-relative render space	14
03.07	Scene object and component model	20
03.08	Scene loading and instantiation	12
03.09	Spatial data structures	16
03.10	World cell partitioning	16
03.11	Asynchronous world streaming	20
03.12	World state persistence	10
03.13	Editor world tooling contracts	8
03.14	Precision and streaming test programme	8

Chapter detail budget totals 200 pages. The remaining -20 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 03 detailed chapter inventory

03.01 Mathematics conventions [12 pages]

Coverage: handedness; axes; angles; matrix storage; precision rules.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.02 Vectors, matrices and quaternions [16 pages]

Coverage: SIMD boundary; normalization; robust comparisons; tests.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.03 Transforms and hierarchy propagation [18 pages]

Coverage: local/world transforms; dirty propagation; cycles; batch updates.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.04 Bounds and geometric primitives [12 pages]

Coverage: AABB; OBB; spheres; planes; frusta; rays.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.05 Double-precision world coordinates [18 pages]

Coverage: WorldPosition; cell+offset; serialization; origin-independent math.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.06 Camera-relative render space [14 pages]

Coverage: conversion contracts; precision budgets; GPU matrices; debugging.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.07 Scene object and component model [20 pages]

Coverage: UUIDs; component storage; activation; tags; layers; events.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.08 Scene loading and instantiation [12 pages]

Coverage: staging; reference resolution; activation phases; failure rollback.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.09 Spatial data structures [16 pages]

Coverage: BVH; loose octree; grids; update costs; query APIs.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.10 World cell partitioning [16 pages]

Coverage: cell IDs; boundaries; persistent layer; cross-cell references.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.11 Asynchronous world streaming [20 pages]

Coverage: request graph; priorities; cancellation; residency; handoff to rendering/physics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.12 World state persistence [10 pages]

Coverage: runtime state; save regions; dirty tracking; versioning.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.13 Editor world tooling contracts [8 pages]

Coverage: cell visibility; locking; streaming preview; large-world navigation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

03.14 Precision and streaming test programme [8 pages]

Coverage: extreme coordinates; seams; teleports; load storms; determinism.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 03 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 03 consumes approved contracts from earlier volumes, especially Volume 02 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 04 - RHI, Vulkan and Direct3D 12

Shared explicit Render Hardware Interface, GPU resources, synchronization, Vulkan backend, Direct3D 12 backend and conformance.

VOLUME BUDGET

215 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V04.

Volume outcomes

Lock the public contracts and implementation order for rhi, vulkan and direct3d 12.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
04.01	RHI design principles	12
04.02	Adapters, devices and queues	14
04.03	Resource descriptors and handles	18
04.04	Command allocators and command lists	16
04.05	Resource states and barriers	18
04.06	Descriptors and binding model	18
04.07	Pipeline and shader objects	16
04.08	Swap chains and presentation	12
04.09	GPU memory architecture	18
04.10	Synchronization and frame contexts	16
04.11	Vulkan backend bootstrap	16
04.12	Vulkan resource and command implementation	22
04.13	Direct3D 12 backend bootstrap	16
04.14	Direct3D 12 resource and command implementation	22
04.15	RHI conformance and backend parity	15

Chapter detail budget totals 249 pages. The remaining -34 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 04 detailed chapter inventory

04.01 RHI design principles [12 pages]

Coverage: explicit semantics; capabilities; backend neutrality; no D3D11-shaped API.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.02 Adapters, devices and queues [14 pages]

Coverage: selection; feature tiers; queue families; device removal.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.03 Resource descriptors and handles [18 pages]

Coverage: buffers; textures; views; samplers; lifetimes; debug names.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.04 Command allocators and command lists [16 pages]

Coverage: recording model; threading; reset rules; submission.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.05 Resource states and barriers [18 pages]

Coverage: state model; transitions; ownership transfer; validation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.06 Descriptors and binding model [18 pages]

Coverage: bindless direction; descriptor heaps/sets; updates; limits.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.07 Pipeline and shader objects [16 pages]

Coverage: graphics; compute; pipeline cache; specialization.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.08 Swap chains and presentation [12 pages]

Coverage: formats; HDR; resize; tearing; frame pacing.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.09 GPU memory architecture [18 pages]

Coverage: heaps; suballocation; upload/readback; budget queries; defrag policy.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.10 Synchronization and frame contexts [16 pages]

Coverage: fences; semaphores; timeline values; frames in flight.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.11 Vulkan backend bootstrap [16 pages]

Coverage: instance; validation; device; extensions; surface.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.12 Vulkan resource and command implementation [22 pages]

Coverage: VMA decision; descriptors; pipelines; barriers; debug utils.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.13 Direct3D 12 backend bootstrap [16 pages]

Coverage: factory; adapter; device; debug layer; DRED.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.14 Direct3D 12 resource and command implementation [22 pages]

Coverage: heaps; root signatures; PSOs; barriers; PIX markers.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

04.15 RHI conformance and backend parity [15 pages]

Coverage: shared tests; capture tests; capability matrix; performance baseline.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 04 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 04 consumes approved contracts from earlier volumes, especially Volume 03 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 05 - Forward+, Deferred Rendering and Render Features

Render graph, scene extraction, visibility, clustered Forward+, deferred G-buffer path, shadows, post-processing and presentation.

VOLUME BUDGET

225 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V05.

Volume outcomes

Lock the public contracts and implementation order for forward+, deferred rendering and render features.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
05.01	Renderer module boundaries	10
05.02	Render graph	22
05.03	Scene extraction and render proxies	16
05.04	Visibility and culling	18
05.05	Lighting data model	14
05.06	Cluster construction and light assignment	18
05.07	Forward+ opaque path	18
05.08	Deferred G-buffer design	18
05.09	Deferred lighting pass	16
05.10	Transparency and special forward passes	14
05.11	Shadow architecture	22
05.12	Sky, atmosphere and fog	12
05.13	Post-processing stack	20
05.14	Editor and debug rendering	10

ID	CHAPTER	PAGES
05.15	Renderer performance and image tests	13

Chapter detail budget totals 241 pages. The remaining -16 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 05 detailed chapter inventory

05.01 Renderer module boundaries [10 pages]

Coverage: scene/render split; frame data; thread ownership; plugin points.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.02 Render graph [22 pages]

Coverage: pass declarations; resource lifetimes; barriers; aliasing; debug viewer.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.03 Scene extraction and render proxies [16 pages]

Coverage: double buffering; immutable frame packets; lifetime; large-world conversion.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.04 Visibility and culling [18 pages]

Coverage: frustum; occlusion; LOD; instancing; GPU-driven path.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.05 Lighting data model [14 pages]

Coverage: light types; units; layers; cookies; probes.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.06 Cluster construction and light assignment [18 pages]

Coverage: 3D clusters; depth slicing; overflow; GPU kernels; debug.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.07 Forward+ opaque path [18 pages]

Coverage: depth prepass; material sorting; MSAA; special materials.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.08 Deferred G-buffer design [18 pages]

Coverage: layout; formats; bandwidth; material IDs; motion vectors.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.09 Deferred lighting pass [16 pages]

Coverage: light volumes; cluster reuse; IBL; decals; probes.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.10 Transparency and special forward passes [14 pages]

Coverage: sorting; OIT deferral; particles; hair-like materials.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.11 Shadow architecture [22 pages]

Coverage: directional cascades; spot; point; atlases; cache; bias.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.12 Sky, atmosphere and fog [12 pages]

Coverage: sky model; environment maps; height fog; volumetrics deferred.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.13 Post-processing stack [20 pages]

Coverage: exposure; tone mapping; bloom; TAA; FXAA; SSAO; DOF; motion blur.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.14 Editor and debug rendering [10 pages]

Coverage: selection; gizmos; wireframe; buffer views; overdraw.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

05.15 Renderer performance and image tests [13 pages]

Coverage: golden scenes; frame budgets; GPU timing; cross-backend parity.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 05 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 05 consumes approved contracts from earlier volumes, especially Volume 04 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 06 - Shaders, Materials, Assets and Content Pipeline

Shader toolchain, material system, asset registry, importers, derived data, cooking, packaging and editor asset workflows.

VOLUME BUDGET

175 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V06.

Volume outcomes

- Lock the public contracts and implementation order for shaders, materials, assets and content pipeline.
- Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.
- Define explicit completion gates consumed by dependent volumes.
- Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
06.01	Shader source policy and compiler pipeline	18
06.02	Shader reflection and binding metadata	14
06.03	Permutation and pipeline cache strategy	16
06.04	Material domain model	16
06.05	Standard PBR material	14
06.06	Advanced shader-backed materials	12
06.07	Asset identity and registry	16
06.08	Import pipeline architecture	18
06.09	Texture import and processing	14
06.10	Mesh and skeleton import	18
06.11	Audio and miscellaneous importers	10
06.12	Derived Data Cache	14
06.13	Cooking and packaging	16
06.14	Asset browser and diagnostics	10

ID	CHAPTER	PAGES
06.15	Pipeline tests and acceptance	9

Chapter detail budget totals 215 pages. The remaining -40 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 06 detailed chapter inventory

06.01 Shader source policy and compiler pipeline [18 pages]

Coverage: HLSL; DXC; DXIL; SPIR-V; include graph; diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.02 Shader reflection and binding metadata [14 pages]

Coverage: resources; constants; vertex layouts; validation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.03 Permutation and pipeline cache strategy [16 pages]

Coverage: feature keys; variant pruning; warmup; disk cache.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.04 Material domain model [16 pages]

Coverage: surface domains; blend modes; shading models; two-sided.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.05 Standard PBR material [14 pages]

Coverage: parameters; textures; defaults; packing; instances.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.06 Advanced shader-backed materials [12 pages]

Coverage: custom shaders; safe extension; fallbacks; editor metadata.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.07 Asset identity and registry [16 pages]

Coverage: UUIDs; paths; redirectors; dependencies; tags.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.08 Import pipeline architecture [18 pages]

Coverage: source files; import settings; workers; cancellation; reimport.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.09 Texture import and processing [14 pages]

Coverage: formats; mips; compression; colour space; virtual textures deferred.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.10 Mesh and skeleton import [18 pages]

Coverage: glTF; coordinate conversion; LODs; skin; collision.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.11 Audio and miscellaneous importers [10 pages]

Coverage: WAV; fonts; localization tables; plugin importers.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.12 Derived Data Cache [14 pages]

Coverage: keys; local cache; invalidation; cleanup; shared cache future.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.13 Cooking and packaging [16 pages]

Coverage: target data; archives; incremental cook; integrity; shipping layout.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.14 Asset browser and diagnostics [10 pages]

Coverage: thumbnails; reference viewer; missing assets; audit.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

06.15 Pipeline tests and acceptance [9 pages]

Coverage: round trips; determinism; corruption; performance; licensing.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 06 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 06 consumes approved contracts from earlier volumes, especially Volume 05 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 07 - Physics, Animation, Audio, Input and Navigation

Jolt integration, body components, character control, soft bodies, animation, audio, input, navigation and lightweight AI utilities.

VOLUME BUDGET

190 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V07.

Volume outcomes

- Lock the public contracts and implementation order for physics, animation, audio, input and navigation.
- Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.
- Define explicit completion gates consumed by dependent volumes.
- Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
07.01	Simulation scheduling and fixed timestep	12
07.02	Jolt integration boundary	14
07.03	Collision shapes and materials	14
07.04	StaticBody3D and RigidBody3D	18
07.05	CharacterBody3D and controller	16
07.06	SoftBody3D	18
07.07	Queries, triggers and constraints	16
07.08	Large-world physics strategy prototype	12
07.09	Animation data and skeleton runtime	16
07.10	Animation graphs and state machines	18
07.11	Audio architecture	14
07.12	Input system	12
07.13	Navigation mesh and agents	14
07.14	Lightweight AI services	8

ID	CHAPTER	PAGES
07.15	Simulation test and performance gate	8

Chapter detail budget totals 210 pages. The remaining -20 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 07 detailed chapter inventory

07.01 Simulation scheduling and fixed timestep [12 pages]

Coverage: tick order; substeps; interpolation; pause; determinism.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.02 Jolt integration boundary [14 pages]

Coverage: third-party isolation; allocators; jobs; coordinate conversion; debug.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.03 Collision shapes and materials [14 pages]

Coverage: primitives; convex; mesh; heightfield; compound; cooking.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.04 StaticBody3D and RigidBody3D [18 pages]

Coverage: components; mass; sleeping; CCD; events; teleports.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.05 CharacterBody3D and controller [16 pages]

Coverage: grounding; steps; slopes; moving platforms; network hooks.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.06 SoftBody3D [18 pages]

Coverage: assets; constraints; attachments; render mesh coupling; budgets.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.07 Queries, triggers and constraints [16 pages]

Coverage: ray/shape casts; overlaps; events; joints; ragdolls.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.08 Large-world physics strategy prototype [12 pages]

Coverage: rebasing; regions; handoffs; precision tests.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.09 Animation data and skeleton runtime [16 pages]

Coverage: poses; clips; compression; sampling; retargeting.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.10 Animation graphs and state machines [18 pages]

Coverage: blend trees; layers; masks; events; root motion; IK.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.11 Audio architecture [14 pages]

Coverage: backend; voices; buses; 3D spatialization; streaming; large worlds.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.12 Input system [12 pages]

Coverage: devices; actions; contexts; rebinding; gamepad; editor separation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.13 Navigation mesh and agents [14 pages]

Coverage: baking; tiles; queries; obstacles; streaming.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.14 Lightweight AI services [8 pages]

Coverage: blackboards; state machines; perception hooks; spatial queries.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

07.15 Simulation test and performance gate [8 pages]

Coverage: stress scenes; determinism; latency; memory; profiling.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 07 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 07 consumes approved contracts from earlier volumes, especially Volume 06 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 08 - SkeinScript and Plugin Architecture

SkeinScript language, compiler, VM, debugging, native bindings, plugin manifests, lifecycle, versioning and extension points.

VOLUME BUDGET

155 planned pages across 14 chapters. Document code: SKEIN-IMP-002-V08.

Volume outcomes

Lock the public contracts and implementation order for skeinscript and plugin architecture.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
08.01	Language goals and grammar boundary	10
08.02	Lexer and parser	14
08.03	Semantic analysis and type system	16
08.04	Bytecode format and verifier	14
08.05	Virtual machine	18
08.06	Managed lifetime model	14
08.07	Native C++ binding layer	16
08.08	Events, coroutines and tasks	14
08.09	Hot reload and state migration	12
08.10	SkeinScript debugger and editor services	14
08.11	Plugin manifest and packaging	10
08.12	Plugin load/unload lifecycle	12
08.13	Plugin extension surfaces	10
08.14	Compatibility, security and test gate	11

Chapter detail budget totals 185 pages. The remaining -30 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 08 detailed chapter inventory

08.01 Language goals and grammar boundary [10 pages]

Coverage: syntax; types; modules; small-language rule; compatibility.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.02 Lexer and parser [14 pages]

Coverage: tokens; AST; error recovery; source maps; tests.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.03 Semantic analysis and type system [16 pages]

Coverage: symbols; overloads; conversion; inheritance; diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.04 Bytecode format and verifier [14 pages]

Coverage: instructions; constants; versions; validation; disassembly.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.05 Virtual machine [18 pages]

Coverage: frames; calls; exceptions/errors; dispatch; performance.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.06 Managed lifetime model [14 pages]

Coverage: GC/reference counting decision; roots; native handles; cycles; budgets.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.07 Native C++ binding layer [16 pages]

Coverage: reflection bridge; marshallng; ownership; thread rules; generated bindings.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.08 Events, coroutines and tasks [14 pages]

Coverage: yielding; engine jobs; cancellation; lifetime; debugging.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.09 Hot reload and state migration [12 pages]

Coverage: compile pipeline; class replacement; instance state; failure rollback.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.10 SkeinScript debugger and editor services [14 pages]

Coverage: breakpoints; stepping; inspection; autocomplete; documentation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.11 Plugin manifest and packaging [10 pages]

Coverage: identity; version; dependencies; content; platforms.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.12 Plugin load/unload lifecycle [12 pages]

Coverage: phases; registration; safe unload; hot reload; error containment.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.13 Plugin extension surfaces [10 pages]

Coverage: components; assets; editor; render passes; script bindings; build steps.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

08.14 Compatibility, security and test gate [11 pages]

Coverage: API versioning; sandboxing; fuzzing; plugin matrix; performance.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 08 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 08 consumes approved contracts from earlier volumes, especially Volume 07 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 09 - SkeinUI, SkeinEditor and Project Framework

Retained native UI, rendering, text, input, docking, editor shell, inspectors, viewports, project model and game-facing UI.

VOLUME BUDGET

210 planned pages across 18 chapters. Document code: SKEIN-IMP-002-V09.

Volume outcomes

Lock the public contracts and implementation order for skeinui, skeineditor and project framework.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
09.01	SkeinUI architecture and widget lifetime	14
09.02	Declarative C++ builder API	14
09.03	Layout and geometry	18
09.04	Invalidation and update model	14
09.05	Input, focus and commands	18
09.06	Text and font system	18
09.07	UI rendering and batching	18
09.08	Standard widget library	20
09.09	Styling and themes	12
09.10	Docking, windows and workspaces	16
09.11	SkeinEditor application shell	14
09.12	Hierarchy, Inspector and transactions	18
09.13	3D viewport and editor interaction	20
09.14	Asset browser and specialist editors	14

ID	CHAPTER	PAGES
09.15	World, navigation and physics tools	12
09.16	Project and game framework	14
09.17	Game UI and SkeinScript integration	10
09.18	UI/editor diagnostics and acceptance	8

Chapter detail budget totals 272 pages. The remaining -62 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 09 detailed chapter inventory

09.01 SkeinUI architecture and widget lifetime [14 pages]

Coverage: retained tree; ownership; UI thread; weak references; composition.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.02 Declarative C++ builder API [14 pages]

Coverage: builders; slots; properties; callbacks; source diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.03 Layout and geometry [18 pages]

Coverage: desired size; arrange; stacks; grids; overlays; DPI; clipping.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.04 Invalidation and update model [14 pages]

Coverage: layout/paint invalidation; sleeping; volatile widgets; timers.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.05 Input, focus and commands [18 pages]

Coverage: routing; capture; keyboard focus; gamepad navigation; shortcuts; commands.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.06 Text and font system [18 pages]

Coverage: FreeType/HarfBuzz review; shaping; fallback; selection; rich text; atlases.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.07 UI rendering and batching [18 pages]

Coverage: draw elements; nine-slice; clips; transient buffers; render graph.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.08 Standard widget library [20 pages]

Coverage: controls; menus; toolbars; dialogs; tables; trees; virtualization.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.09 Styling and themes [12 pages]

Coverage: style registry; dark/light/high contrast; plugin styles; project overrides.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.10 Docking, windows and workspaces [16 pages]

Coverage: native windows; tabs; splits; floating; multi-monitor; persistence.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.11 SkeinEditor application shell [14 pages]

Coverage: startup; project manager; commands; menus; crash recovery.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.12 Hierarchy, Inspector and transactions [18 pages]

Coverage: scene tree; property editors; multi-edit; undo/redo; plugin inspectors.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.13 3D viewport and editor interaction [20 pages]

Coverage: camera; gizmos; selection; snapping; play modes; debug overlays.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.14 Asset browser and specialist editors [14 pages]

Coverage: search; thumbnails; dependencies; asset editors; import settings.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.15 World, navigation and physics tools [12 pages]

Coverage: cell editor; streaming preview; baking; collision views; terrain baseline.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.16 Project and game framework [14 pages]

Coverage: project descriptor; modules; settings; startup map; game services.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.17 Game UI and SkeinScript integration [10 pages]

Coverage: screens; HUD layers; bindings; gamepad; world-space UI.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

09.18 UI/editor diagnostics and acceptance [8 pages]

Coverage: widget reflector; layout debug; automation tests; performance.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 09 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 09 consumes approved contracts from earlier volumes, especially Volume 08 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 10 - Networking and Starcore Server Architecture

Transport, messaging, serialization, reliability, security hooks, headless runtime, replication foundations and future Linux server path.

VOLUME BUDGET

145 planned pages across 13 chapters. Document code: SKEIN-IMP-002-V10.

Volume outcomes

Lock the public contracts and implementation order for networking and starcore server architecture.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
10.01	Networking goals and authority model	10
10.02	Socket platform abstraction	12
10.03	Packet and message serialization	14
10.04	Connection lifecycle	12
10.05	Reliability and ordering	14
10.06	Time synchronization and clocks	10
10.07	Security and authentication hooks	12
10.08	Headless runtime architecture	14
10.09	Starcore server plugin boundary	14
10.10	Replication foundations	16
10.11	Prediction and reconciliation plan	10
10.12	Network Insights integration	8
10.13	Load, soak and failure testing	9

Chapter detail budget totals 155 pages. The remaining -10 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 10 detailed chapter inventory

10.01 Networking goals and authority model [10 pages]

Coverage: client/server; trust boundaries; tick rates; scope limits.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.02 Socket platform abstraction [12 pages]

Coverage: UDP; TCP; nonblocking I/O; addressing; Windows/Linux boundary.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.03 Packet and message serialization [14 pages]

Coverage: bit streams; schemas; versioning; limits; validation.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.04 Connection lifecycle [12 pages]

Coverage: handshake; timeouts; disconnects; reconnect; diagnostics.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.05 Reliability and ordering [14 pages]

Coverage: sequence numbers; acks; resends; channels; congestion hooks.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.06 Time synchronization and clocks [10 pages]

Coverage: RTT; offset; drift; server tick mapping.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.07 Security and authentication hooks [12 pages]

Coverage: encryption boundary; tokens; rate limits; malformed traffic.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.08 Headless runtime architecture [14 pages]

Coverage: no-render startup; service lifecycle; console; configuration; signals.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.09 Starcore server plugin boundary [14 pages]

Coverage: services; world ownership; persistence hooks; observability.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.10 Replication foundations [16 pages]

Coverage: entity IDs; snapshots; interest management; property descriptors.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.11 Prediction and reconciliation plan [10 pages]

Coverage: input commands; client prediction; corrections; physics boundary.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.12 Network Insights integration [8 pages]

Coverage: packets; messages; bandwidth; latency; loss.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

10.13 Load, soak and failure testing [9 pages]

Coverage: bots; packet loss; long runs; resource limits; recovery.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 10 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 10 consumes approved contracts from earlier volumes, especially Volume 09 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 11 - SkeinInsights, Diagnostics and Profiling

SkeinTrace, trace transport and storage, analysis providers, timing views, memory, jobs, rendering, world streaming and network analysis.

VOLUME BUDGET

135 planned pages across 15 chapters. Document code: SKEIN-IMP-002-V11.

Volume outcomes

Lock the public contracts and implementation order for skeininsights, diagnostics and profiling.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
11.01	Observability architecture	10
11.02	Trace event schema	12
11.03	Per-thread trace buffers	14
11.04	Trace channels and presets	8
11.05	Transport, TraceServer and files	14
11.06	Analysis framework and providers	12
11.07	Timing and frame analysis	14
11.08	Job-system analysis	10
11.09	Memory analysis	14
11.10	Rendering and GPU analysis	10
11.11	World streaming and asset analysis	8
11.12	Network and Starcore analysis	8
11.13	SkeinInsights application UI	12
11.14	Triggered and circular capture	8

ID	CHAPTER	PAGES
11.15	Overhead and correctness gate	5

Chapter detail budget totals 159 pages. The remaining -24 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 11 detailed chapter inventory

11.01 Observability architecture [10 pages]

Coverage: trace/analysis/UI separation; overhead budgets; shipping policy.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.02 Trace event schema [12 pages]

Coverage: event IDs; fields; metadata; self-description; versions.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.03 Per-thread trace buffers [14 pages]

Coverage: lock avoidance; buffering; timestamps; loss accounting; flush.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.04 Trace channels and presets [8 pages]

Coverage: channels; runtime toggles; command line; project presets.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.05 Transport, TraceServer and files [14 pages]

Coverage: local/remote; framing; storage; rotation; recovery.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.06 Analysis framework and providers [12 pages]

Coverage: session model; indexes; queries; extensibility.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.07 Timing and frame analysis [14 pages]

Coverage: CPU lanes; frames; scopes; aggregation; critical path.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.08 Job-system analysis [10 pages]

Coverage: queues; waits; steals; utilisation; dependency graph.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.09 Memory analysis [14 pages]

Coverage: alloc/free; tags; call stacks; lifetimes; leaks; comparison.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.10 Rendering and GPU analysis [10 pages]

Coverage: render graph; queues; draws; resources; frame pacing.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.11 World streaming and asset analysis [8 pages]

Coverage: cells; I/O; residency; dependencies; stalls.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.12 Network and Starcore analysis [8 pages]

Coverage: connections; messages; bandwidth; RTT; server markers.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.13 SkeinInsights application UI [12 pages]

Coverage: session browser; live capture; views; filters; exports.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.14 Triggered and circular capture [8 pages]

Coverage: hitch triggers; assert/crash capture; headless capture.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

11.15 Overhead and correctness gate [5 pages]

Coverage: benchmarks; lost events; clock tests; large traces.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 11 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 11 consumes approved contracts from earlier volumes, especially Volume 10 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

Volume 12 - Build, Testing, Deployment and Governance

Test architecture, CI, performance gates, security, packaging, documentation control, release policy and long-term technical governance.

VOLUME BUDGET

150 planned pages across 14 chapters. Document code: SKEIN-IMP-002-V12.

Volume outcomes

Lock the public contracts and implementation order for build, testing, deployment and governance.

Provide enough API, data, ownership, threading and testing detail to implement without inventing architecture mid-task.

Define explicit completion gates consumed by dependent volumes.

Record deferred work so it cannot quietly re-enter the active milestone.

Chapter budget

ID	CHAPTER	PAGES
12.01	Verification strategy	10
12.02	Unit-test framework and conventions	12
12.03	Integration and subsystem tests	14
12.04	Rendering image tests	12
12.05	Performance benchmark programme	14
12.06	Memory and long-running stability tests	12
12.07	Continuous integration matrix	14
12.08	Static analysis and sanitizers	10
12.09	Security and robustness	12
12.10	Cooking, packaging and deployment	12
12.11	Versioning and compatibility policy	10
12.12	Documentation system	10
12.13	Change control and governance	10
12.14	Release readiness and acceptance	12

Chapter detail budget totals 164 pages. The remaining -14 pages are reserved for cover, control, volume-level dependency maps, appendices and index material.

Volume 12 detailed chapter inventory

12.01 Verification strategy [10 pages]

Coverage: test pyramid; evidence; ownership; failure triage.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.02 Unit-test framework and conventions [12 pages]

Coverage: test discovery; fixtures; assertions; determinism; timing.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.03 Integration and subsystem tests [14 pages]

Coverage: module harnesses; temporary projects; failure injection; cleanup.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.04 Rendering image tests [12 pages]

Coverage: golden images; tolerances; backend matrix; artifact review.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.05 Performance benchmark programme [14 pages]

Coverage: microbenchmarks; scene benchmarks; budgets; regression thresholds.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.06 Memory and long-running stability tests [12 pages]

Coverage: budgets; leaks; fragmentation; soak tests; shutdown.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.07 Continuous integration matrix [14 pages]

Coverage: four presets; clean builds; tests; artifacts; caches.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.08 Static analysis and sanitizers [10 pages]

Coverage: clang-tidy; MSVC analysis; ASan where viable; warnings.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.09 Security and robustness [12 pages]

Coverage: input limits; fuzzing; untrusted assets; plugins; network.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.10 Cooking, packaging and deployment [12 pages]

Coverage: client; editor; server; symbols; manifests; install layouts.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.11 Versioning and compatibility policy [10 pages]

Coverage: engine versions; plugin API; assets; scripts; network protocols.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.12 Documentation system [10 pages]

Coverage: master spec; construction volumes; ADRs; API docs; changelogs.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.13 Change control and governance [10 pages]

Coverage: scope changes; review gates; deprecations; decision records.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

12.14 Release readiness and acceptance [12 pages]

Coverage: release checklist; known issues; rollback; support bundle.

Required treatment: public contract, private structure, ownership/lifetime, threading, allocation behaviour, diagnostics, implementation sequence, automated tests, performance evidence and completion gate.

Volume 12 dependency and acceptance map

Entry conditions

All prerequisite volume chapters referenced by the first implementation milestone are approved.

The four CMake presets configure cleanly from empty build directories.

Public module boundaries are represented in the repository before implementation expands.

Required third-party licences and versions are recorded.

A SkeinInsights trace category and baseline test strategy are identified where applicable.

Exit conditions

All normative public contracts compile under both compilers.

Failure paths, cancellation and shutdown have automated coverage.

Memory and performance budgets have measurements, not guesses.

SkeinInsights or equivalent diagnostics expose the subsystem behaviour required for debugging.

The volume completion checklist is signed off and dependent chapters may begin.

Volume-specific dependency notes

Volume 12 consumes approved contracts from earlier volumes, especially Volume 11 where directly adjacent.

Cross-volume changes must update both affected volume plans and the master dependency register.

4. Cross-volume dependency matrix

VOL	REQUIRES	UNLOCKS
01	None	All volumes
02	01	03, 06, 08, 09, 10
03	01, 02	05, 07, 09, 10
04	01	05, 06, 09, 11
05	03, 04	06, 07, 09, 11
06	02, 04, 05	07, 08, 09, 12
07	01, 03, 05, 06	09, 10, 11
08	01, 02, 06	09, 10
09	02, 03, 05, 06, 08	11
10	01, 02, 03, 07, 08	11, 12
11	01 plus instrumentation from all	12 and every performance gate
12	01 and progressively all volumes	Release readiness

4.1 Parallel authoring rule

Volumes may be drafted in parallel, but normative API sections cannot be finalized ahead of their prerequisite contracts. Parallel drafts must mark unresolved dependencies explicitly and may not invent substitute interfaces merely to appear complete.

5. Milestone map

ID	MILESTONE	EVIDENCE	VOLUMES
M0	Repository baseline	Four presets build and test. Git, README, .gitignore and Command Deck are current.	01, 12
M1	Foundation runtime	Foundation, platform, memory, jobs, lifecycle, logging and basic trace emission.	01, early 11
M2	Data model	Reflection generation, object handles, serialization, migrations and transactions.	02
M3	World core	Scene/component runtime, large coordinates, spatial queries and streaming skeleton.	03
M4	GPU foundation	RHI conformance with Vulkan and D3D12 clearing and presenting frames.	04
M5	Usable renderer	Render graph, Forward+, deferred opaque path, shadows and core post-processing.	05
M6	Content pipeline	Shaders, materials, asset imports, DDC, cooking and package loading.	06
M7	Playable simulation	Jolt bodies, character, animation, audio, input and navigation.	07
M8	Extension layer	SkeinScript VM, native bindings and plugin lifecycle.	08
M9	Tooling environment	SkeinUI, SkeinEditor shell, viewport, inspector and asset browser.	09
M10	Connected runtime	Networking foundation and Starcore headless server plugin path.	10
M11	Deep observability	Full SkeinInsights application and subsystem analysis providers.	11

M12

Release discipline

Complete CI, packaging,
security, performance and
governance gates.

12

6. Document control and numbering

Document set code: SKEIN-IMP-002.

Master plan code: SKEIN-IMP-002-MASTER.

Volume code: SKEIN-IMP-002-VNN, where NN is 01 through 12.

Volume filename: SKEIN-IMP-002_Engine_Construction_Specification_Volume_NN_<Title>_vX.Y.pdf.

Chapter identifiers remain stable after approval. Removed chapters are retired, not renumbered into silence.

Volume v0.x denotes working construction specifications. v1.0 denotes an approved implementation baseline.

Every revision includes a change log and identifies source master-specification changes.

6.1 Review gates

GATE	REQUIREMENT
Outline gate	Chapter inventory and page budget approved.
Contract gate	Public APIs, ownership and dependencies approved before private implementation.
Implementation gate	Ordered work packages are complete and compile.
Verification gate	Automated tests, profiling and failure evidence pass.
Baseline gate	Volume is promoted to an approved construction baseline.

6.2 Immediate authoring order

Approve this master plan.

Author Volume 01 in full, beginning with repository/toolchain and ending with runtime acceptance.

Keep early SkeinTrace hooks synchronized with the future Volume 11 contract.

Create Volume 02 only after Volume 01 public foundation types and module rules are stable.

Review page totals after Volume 01 to calibrate the remaining budget without reducing required coverage.

**NEXT
DELIVERABLE**

SKEIN-IMP-002 Volume 01 - Core Systems and Runtime Foundation, targeted at approximately 240 substantive pages.